SIMATS ENGINEERING

CLASSROOM ACTIVITY – SHORT ANSWER TEST

Topic: Classification of Data Mining Systems

Course	CLCSA16 – Data Warehousing and Data Mining (DWDM)
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Time / Questions	30 minutes | 10 short-answer questions

Q1. Define a Data Mining System. *[BTL1 – Remember]*

A data mining system is a software system that automatically analyses large volumes of data held in databases, data warehouses or other repositories and extracts useful, previously unknown patterns and knowledge from it. It integrates techniques from databases, statistics, machine learning and visualisation to turn raw data into decisions.

Q2. List the different classifications of Data Mining Systems. *[BTL1 – Remember]*

- According to the **kind of data (database) mined**
- According to the **kind of knowledge mined**
- According to the **kind of techniques utilised**
- According to the **application domain adapted**
- According to the **degree of user interaction** (usually treated as part of the technique-based view)

Q3. Explain the classification of Data Mining Systems based on the type of data mined. *[BTL2 – Understand]*

Here a system is named after the data it can handle. A relational mining system works on tables, a warehouse mining system on multidimensional cubes, and a transactional system on purchase or log records. Others are built for complex data: spatial (maps, GPS), time-series (stock prices, ECG), text, multimedia (images, audio, video), web and data-stream mining systems. A system built for one type usually cannot mine another without modification.

Q4. Describe the classification of Data Mining Systems based on the type of knowledge mined. *[BTL2 – Understand]*

Systems are grouped by the mining task (function) they perform:

- **Characterisation and discrimination** – summarise a target class or compare classes.
- **Association and correlation analysis** – find items that occur together.
- **Classification and prediction** – build models that assign labels or forecast values.

- **Clustering** – group similar objects without predefined labels.
- **Outlier and evolution analysis** – detect exceptions and study changes over time.

They can also be separated by the level at which knowledge is found: generalised (high-level), primitive (raw-level) or multilevel.

Q5. Identify the techniques used in Data Mining Systems. *[BTL1 – Remember]*

- Statistics and probability-based methods (regression, Bayesian methods)
- Machine learning – decision trees, k-nearest neighbour, support vector machines, rule induction
- Neural networks and genetic algorithms
- Association-rule mining (Apriori, FP-growth) and clustering (k-means, hierarchical)
- Database- and warehouse-oriented methods such as OLAP and data cubes
- Pattern recognition, visualisation and high-performance (parallel) computing

Q6. Explain the classification of Data Mining Systems based on applications. *[BTL2 – Understand]*

Systems are tailored to the needs of a particular industry, so the patterns and data they target differ. Typical examples are:

Application domain	Typical use of the mining system
Banking and finance	Credit scoring, fraud detection, loan default prediction
Retail and e-commerce	Market-basket analysis, recommendations, customer segmentation
Telecommunications	Churn prediction, call-pattern and network-fault analysis
Healthcare and biology	Disease prediction, gene and drug analysis

Domain-specific systems usually perform better than generic ones because they embed domain knowledge.

Q7. Describe the classification of Data Mining Systems based on user interaction. *[BTL2 – Understand]*

- **Autonomous systems** – run on their own with little or no user input and report the discovered patterns.
- **Interactive exploratory systems** – the user steers the mining by choosing data, drilling down and refining results step by step.
- **Query-driven systems** – the user states the task in a mining query language (such as DMQL) and the system answers that specific request.

Q8. Illustrate a real-world application of a Data Mining System in healthcare. *[BTL3 – Apply]*

Diabetes risk prediction in a hospital. The hospital collects patient records containing age, BMI, blood pressure, glucose level and family history. After cleaning the data, a classification model such as a decision tree is trained on

past patients whose outcome (diabetic / non-diabetic) is known. For a new patient the system predicts the risk class, so doctors can recommend early screening and lifestyle changes.

The benefit is earlier diagnosis, lower treatment cost and better use of hospital resources.

Q9. Define classification in data mining. *[BTL1 – Remember]*

Classification is a supervised learning technique in which a model (classifier) is built from training data whose class labels are already known, and is then used to predict the class label of new, unseen records. It works in two steps: learning the model, then using it for prediction. Example: labelling an e-mail as spam or not spam.

Q10. Differentiate between classification and clustering in data mining. *[BTL2 – Understand]*

Basis	Classification	Clustering
Learning type	Supervised	Unsupervised
Class labels	Predefined and available in training data	Not available; groups are discovered
Goal	Predict the class of new objects	Find natural groupings of similar objects
Typical algorithms	Decision tree, Naive Bayes, SVM, k-NN	k-means, hierarchical, DBSCAN
Evaluation	Accuracy, precision, recall	Cluster quality (e.g. silhouette)
Example	Marking a loan applicant as safe or risky	Segmenting customers by buying behaviour